



CIIT Training Institute Pvt Ltd

Dot Net Core Micro Services with Azure DevOps and Gen AI

Course Duration: 4 Months

Total Fees: 25,000+ 18% GST / 28,000 +18% GST

Core Benefits

Specialized training that combines **ASP.NET Core Microservices, DevOps, and AI Integration** positions you at the absolute top tier of the software engineering job market.

While a standard web developer builds basic applications, an engineer trained in this trifecta builds **autonomous, intelligent, self-scaling enterprise ecosystems**. This specific hybrid skillset represents the most lucrative and future-proof career path in the Microsoft ecosystem today.

- **Exponential Salary Premium (The "Multiplier" Effect):** When you combine these three domains, your market value scales exponentially because you solve three of the most expensive problems an enterprise faces: scalability, automated delivery, and intelligent automation.
- **Elimination of Competition via a Rare Skillset:** The tech industry is flooded with developers who can write standard REST APIs. However, very few developers know how to:
 - Containerise a .NET application using Docker and orchestrate it via Kubernetes.
 - Connect those services asynchronously using MassTransit or RabbitMQ.

- Safely embed an AI agent using Semantic Kernel without creating performance bottlenecks or massive cloud cost overruns.
By mastering this intersection, you completely bypass the crowded entry-to-mid-level developer market.

Strategic Career Outcomes

- **Senior .NET Software Engineer / Tech Lead:** You move away from building basic CRUD apps to designing fault-tolerant, event-driven enterprise ecosystems.
- **Cloud-Native / Solutions Architect:** You become the bridge between business requirements and cloud infrastructure, deciding *how* services interact, scale, and stay secure.
- **DevOps / Platform Engineer (.NET Ecosystem):** Your knowledge of Docker, Kubernetes, and automated CI/CD pipelines makes you highly valuable for modern platform engineering teams.

100% Money Back Guarantee

5-Day Money-Back Guarantee: Till date We have trained 10000+ candidates including under graduates, freshers and working professionals with expert knowledge. All are satisfied with our training quality. So we are giving guarantee to all those who want to join our institute that if you're disappointed for whatever reason, you'll get your 100% refund. We won't make you invoke any rules or conditions – if you're not satisfied within your first 5 days then we'll refund you without any excuse.

Syllabus

ASP.NET Core Microservices

- **Architecture Foundations & Monolith Migration**
 - **Monolithic vs. Distributed Systems:** Core trade-offs, advantages, and drawbacks
 - **Microservices Principles:** Single Responsibility, loose coupling, high cohesion, and independent scalability
 - **Deconstruction & Boundaries:** Identifying service boundaries and migrating via the Strangler Fig Pattern

- **System Design Basics:** High-Level Design (HLD), Low-Level Design (LLD), and defining functional/non-functional requirements
- **Data Management & Service Design**
 - **Database-per-Service Pattern:** Ensuring data isolation across distributed databases
 - **Advanced Design Patterns:** Implementing **CQRS** (Command Query Responsibility Segregation) and **Domain-Driven Design (DDD)**
 - **Distributed Caching:** Using **Redis** to boost application response performance
 - **Clean Architecture:** Organizing project code using Clean/Vertical Slice Architecture with Entity Framework Core.
- **Inter-Service Communication**
 - **Synchronous Protocols:** Structuring RESTful Web APIs, handling internal communication via **gRPC**, and managing **IHttpClientFactory**
 - **Asynchronous Messaging (Event-Driven):** Implementing pub/sub models utilizing message brokers like **RabbitMQ**, **Kafka**, or **Azure Service Bus**.
 - **Abstraction Frameworks:** Integrating **MassTransit** to streamline event publishing and consumption.
- **Routing & Edge Infrastructure (API Gateways)**
 - **API Gateway Patterns:** *Request routing, protocol translation, and request aggregation.*
 - **Gateway Technologies:** Hands-on experience configuring Ocelot or Microsoft's reverse proxy YARP.
 - **Cross-Cutting Concerns:** Implementing rate-limiting, request throttling, and unified CORS policies at the edge.
- **Security & Identity Management**
 - **Distributed Authentication:** Token-based security architecture using JWT, OAuth2, and OpenID Connect.
 - **Identity Management:** Integrating centralized auth systems like **ASP.NET Core Identity**.
 - **Role-Based Authorization:** Validating scopes and user roles down to individual microservice endpoints
- **Containerization, Orchestration & DevOps**
 - **Docker Essentials:** Writing highly optimized Dockerfiles, multi-stage builds, and running complex networks via Docker Compose.
 - **Kubernetes (K8s):** Deploying, scaling, and managing pod lifecycles, services, and configuration files.

- **Observability:** Centralized logging, distributed tracing, and health checks.
- **CI/CD Pipelines:** Automating test, build, and deployment structures using GitHub Actions or **Azure DevOps**.

Azure DevOps & Cloud Integration

- **DevOps Fundamentals:** Understanding the DevOps lifecycle, Agile methodologies (Scrum), and the importance of CI/CD.
- **Version Control:** In-depth Git and Azure Repos (branching strategies, merging, pull requests, and command line operations).
- **Azure Pipelines (CI/CD):** Creating and managing automated build and release pipelines for continuous integration and continuous deployment.
- **Azure Boards:** Utilizing boards for project management, backlog management, sprint planning, and tracking work items.
- **Azure Artifacts/Test Plans:** Managing application dependencies and implementing test management strategies.
- **Containerization & Orchestration:**
 - **Docker:** Building and managing Docker images and containers using Docker files.
 - **Kubernetes (AKS):** Introduction to Azure Kubernetes Service for orchestrating containerized applications.
- **Azure Cloud Services:**
 - **Azure Fundamentals:** Core concepts of Azure regions, availability zones, and infrastructure.
 - **Deployment Targets:** Deploying applications to Azure App Services, Azure Functions, and Virtual Machines.
 - **Infrastructure as Code:** Overview of ARM templates or Terraform for automating infrastructure deployment.
 - **Monitoring:** Using Azure Monitor and Application Insights for logging and performance monitoring.

Generative AI (GenAI)

- **Large Language Models (LLMs):** Understanding open vs. closed

models, cost structures, and latency.

- **Prompt Engineering:** Techniques for effective prompting and practice scenarios.
- **Retrieval-Augmented Generation (RAG):** Connecting models to live data using vector databases like Pinecone or ChromaDB.
- **Fine-tuning:** Customising LLMs for specific industry use cases.

Project Work & Interview Preparation

- 2+ Real Time Projects
- Weekly 2 technical mock interviews
- Weekly 2 Programming Tests
- Weekly 2 HR mock interviews

Our Alumni works at Top MNCs & Indian Companies. So Can You!



KEY Features that Make Us Better and Best Campus for You